Telescoping series (7.15)

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(Telescoping series.) Let $Q_n = 1 + 1/4 + 1/9 + \cdots + 1/n^2$. Then, for $n \geq 3$,

$$\frac{19}{12} - \frac{1}{n+1} < Q_n < \frac{7}{4} - \frac{1}{n}.$$